

CANON. — Master Research Dossier

Dynamic HMI generation at runtime + machine control via HMI

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Compiled from a 6-agent / 4-model deep-research swarm (Opus · Sonnet · Haiku · Fable)

Coverage: competitive landscape · standards & machine-context models · prior-art & patents · ROI & market · datasets & data sources · technical build stack

Evidence labelled VERIFIED / STRONG / INFERENCE · September 2026

CANON / PS2 — MASTER RESEARCH DOSSIER

PS2 = Dynamic HMI generation at runtime + machine control via HMI. Team DigiSeva · Schneider HMI Hackathon 2026. Research by a 6-agent, 4-model swarm (Opus/Sonnet/Haiku/Fable). Labels: VERIFIED / STRONG / INFERENCE.

0. THE ONE-PARAGRAPH VERDICT

Exhaustive search (all major vendors + startups + patents + academia, 2024–2026) confirms **no product or paper ships the FULL LOOP**: a *canonical fused machine model* → **runtime, on-demand, validated** HMI generation with **PLC-authoritative guardrails** → **LLM-reasoned change-impact continuity**. Every piece has prior art individually; the integrated loop is **genuine whitespace (verified by absence)**. CANON must therefore NOT pitch "AI generates an HMI" (patented since 2007) — it must pitch "**a generation-plus-guarantee system**": **validated bindings + PLC authority + change-continuity, at runtime**.

1. COMPETITIVE LANDSCAPE (VERIFIED, multi-agent)

Headline (VERIFIED by absence across Siemens, Rockwell, Schneider, AVEVA, Ignition, COPA-DATA, GE, Emerson, Honeywell, ABB, startups): *No vendor ships runtime, generative HMI creation with machine-control authority.* All generative HMI = **build/design-time, human-gated**. Runtime "dynamic" = pre-engineered faceplates/scripts/bindings.

Closest competitors (must differentiate from these by name)

- **Siemens Eigen Engineering Agent** — GA 20 Apr 2026, TIA Portal. Agentic, **self-validating**, generates PLC code + **WinCC Unified HMI** (mostly JS behavior / VB migration, not clearly full NL screen-layout). **BUILD-TIME**, engineer-reviewed, no runtime, no PLC control authority. *The strongest "we already do that" threat.* [VERIFIED]
- **Tatsoft FrameworkX AI Designer** — shipping Mar 2026; **MCP-based** (Claude/Cursor/Copilot drive the config API: tags, alarms, **screens**, historian); claims 40–50% timeline cut. **BUILD-TIME** config generation (+ runtime AI *chat* over live data, on-prem). Architecturally the most direct analog. [VERIFIED]
- **Honeywell "vibe engineering" / Experion Cognition** — preview, 2026; NL → HMI displays + P&IDs + tests, **build-time**, DCS-scale, human-gated sandbox. [STRONG]
- **Siemens SiVArc** — genuine model-driven HMI screen/tag generation **from the PLC program via rules** (NOT AI), build-time. The incumbent deterministic baseline. [VERIFIED]
- **Runtime NL machine control** — research only: *arXiv 2510.11300* "Beyond touch-based HMI" (LLM + OPC UA tool-calling controls machines via NL); **MCP-for-PLC** emerging plumbing (no safety validation, no HMI gen). [STRONG]

Model/template-driven HMI (mature, NON-AI — do not claim as novel)

Siemens WinCC Unified faceplates(UDT) · Rockwell Optix "Types" · AVEVA ArchestrA templates · **Ignition UDT + indirect binding** (runtime *retargeting*, not generation) · **COPA-DATA zenon Smart Objects** (deterministic template instantiation) · GE Proficy model-based HMI · ABB 800xA aspect objects. **NAMUR MTP** auto-generates a module's HMI from its package (standardized, ~70% effort cut) — the closest *standardized* precedent, but static/build-time/process-only.

Reality-check (great pitch ammo)

ARC Advisory "LLM-Wrapper Illusion" (570 decision-makers, 100+ vendors, 40+ copilots): **>65% fail under real operational stress**; most are chat sidecars lacking "cyber-physical grounding." ARC prescribes **"generative UI + hardcoded execution bounds + living knowledge graph"** → *that is exactly CANON's design.* [VERIFIED]

CANON whitespace (defensible): runtime + validated bindings + PLC authority + change-continuity, ingesting MTP/AAS/OPC-UA/UDT.

MUST NOT claim: "novel/first AI-generated HMI"; "nobody generates HMI with AI." **SAY:** differentiation = machine-context-driven generation with validated bindings, controlled interaction, change-aware regeneration — a *guarantee* layer, not a generator.

2. STANDARDS / MACHINE-CONTEXT MODELS (VERIFIED) — credibility backbone

- **ISA-101** (HMI): 4-level hierarchy (L1 overview→L4 diagnostics), restrained color (gray normal, color only for abnormal). → **hard, checkable render rule** in the generator (demoable).
 - **ISA-18.2 / IEC 62682 / EEMUA 191** (alarms): each alarm = {tag, setpoint, priority, class, cause, consequence, required action} in a Master Alarm DB. **Flood = >10/10 min** → live computed state.
 - **ISA-95 / IEC 62264 + ISA-88**: equipment hierarchy Enterprise→Site→Area→Cell→Unit→Equipment Module→Control Module → canonical-model skeleton + HMI nav.
 - **OPC UA + companion specs** (DI, Machinery, PackML, AML): browsable object graph (NodeSet2.xml). → ingest/expose canonical model as OPC UA info model = **"read the type graph → generate the screen"** (best live-demo credibility).
 - **AutomationML / IEC 62714** (CAEX+COLLADA+PLCopenXML): engineering exchange incl. HMI. → **offline ingestion** format.
 - **AAS / IEC 63278** (IDTA): asset digital-twin shell of submodels. **PRECISION: NO official AAS "HMI"/visualization submodel** → use AAS as asset-metadata envelope ONLY.
 - **NAMUR MTP (VDI/VDE/NAMUR 2658 / IEC 63280)** — deepest hit: **Part 2 = HMI, Part 3 = DataAssembly** (typed value+status+unit+limits), **Part 6/7 = alarms**. Frame CANON as **"MTP-inspired."** **Schneider itself publicly presents on MTP** → lands with judges. **MTP 2.0 released 2026.**
 - **PackML/OMAC**: standard machine-state model (17 states) → standard state widget.
- Name-drop top 4:** MTP · OPC UA(+companion) · ISA-18.2/IEC 62682 · ISA-101. **Claim rule:** "aligned to / informed by," NEVER "compliant / certified / conformant."

3. PRIOR-ART / PATENTS (VERIFIED) — the honesty firewall

"AI/model-driven HMI generation" has **15+ years of patents**:

Patent	Assignee	Date	Covers
US8,533,619B2	Rockwell	2007/2013	context/state-driven dynamic HMI generation (ML-inferred intent)
US9,235,454B2	Siemens	2012/2016	server generates HMI on-demand from an engineering database + live state , role/event-based
US11,175,931 / US11,861,379	AVEVA	2016/21-24	runtime auto-compose GUI by traversing a navigation/metadata model ("data-driven HMI")
US20250005224A1	Rockwell	filed 2023, pub Jan 2025 (pending)	LLM generates HMI faceplates "with connections to the respective logic" + response validation — the closest to CANON
US10,281,894 / US8,798,775	(bind graphics to controller data; detect new/deleted PLC tags → add/flag HMI element)	—	deterministic change-reconciliation prior art

MTP = non-patent prior art for "auto-gen HMI from canonical module model." **Honest novelty:** the **compound** {generation FROM canonical single-source model + *validated/typed* bindings + *PLC-as-authority* consistency + *change-continuity* with audit/operator-state retention}. **No**

single prior-art hit combines all four — esp. PLC-authority invariant & continuity appear **unclaimed**. **Do FTO counsel** vs US20250005224A1 & US9,235,454 before any IP claim. **NEVER say**: "first to use AI to generate HMI"; "no one generates HMI from NL"; "context/role-based dynamic HMI gen is novel." (all false).

4. ROI / MARKET / USE CASES (VERIFIED)

- PS's "50–70% HMI lifecycle = screens" has **no primary source** (cite as "per the problem statement"). Back with: **MTP "up to 70% engineering-effort reduction"** (best sourced); **70% of HMI designers hand-script screens** (Pro-face/Schneider); unified platforms **~50%** cut; commissioning cuts **Siemens 70% / Rockwell 60% / Schneider ~50% (500 hrs)**, ARC 15–30%.
 - SCADA market \$12.9B(2025)→\$13.9B(2026)**, **~8.5% CAGR**. Digital twin **\$39.5B→\$53.6B**; 43% use ≥1 twin, 59% plan by 2028. 76% mfrs implementing AI by 2026. Boehm cost-of-change \$1→\$10→\$100. Automation ROI 10:1–30:1, payback 12–18 mo.
 - Use cases (quantified, guarded)**: brownfield migration (200–400 hrs screen re-authoring; ~50% saved) · multi-machine variant scale-out (70–80% cut) · commissioning (target 40–50%) · machine-change adaptability (40–50% change-cycle cut) · size-independent HMI (~80% responsive-layout cut) · operator onboarding (train –20–30%, errors –40–60% per ISA-101).
 - Do NOT claim numerically**: "50–70% lifecycle is screens" as fact; "AI cuts commissioning 70%" (that's Siemens motion VC); universal \$ rework; "90% twin adoption"; universal "genAI cuts eng X%."
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5. DATASETS & DATA SOURCES (VERIFIED free) — buildable now

OPC UA servers (expose machine model as info model): node-opcua sample server ★ (Node.js, 5-min) · open62541 ★ (C, Pi) · Prosys Simulation Server (free) · Unified Automation demo + UaExpert client · digitalpetri Milo (online opc.tcp://milo.digitalpetri.com:62541/milo). **PLC/Modbus sims (control + permissive reject)**: pymodbus ★ · ModbusPal ★ · OpenPLC ★ (IEC 61131-3 runtime, Pi) · CODESYS sim · Factory I/O (3D, trial). **Canonical-model inputs**: AutomationML samples (github.com/AutomationML/AMLEngine2.1, PyAutomationML) · AAS/AASX (IDTA specs, Bosch semantic stack) · MTP sample (VDI/VDE/NAMUR 2658) · ISA-5.1 tag naming (PT/TT/XV...). **Live data for trends/alarms: Tennessee Eastman** ★ (52 vars, 21 faults, Kaggle/IEEE) · **Wind-Turbine SCADA** (Zenodo 10958775) · **Industrial Alarm Monitoring 2018–2024** ★ (Kaggle, 6 yrs) · ICS Alarm Text (Kaggle) · beverage bottling line (Zenodo 18146866). **HMI components/ref**: hmlibrary.com (900+ ISA-101 SVG) · ISA-101 docs · Node-RED PLC-sim flows · Ignition/Kepware demos (benchmark). **Recommended 30-min stack**: node-opcua sample server + pymodbus (Tennessee-Eastman CSV playback) + minimal AML/JSON machine model (pump+3 sensors+2 valves) + Kaggle alarm CSV → CANON reads OPC UA → generates ISA-101 screen.

6. TECHNICAL STACK (VERIFIED build blueprint)

The safety wedge (key idea): PROPOSE → COMPILE → VALIDATE. The LLM only emits a **layout proposal** in a JSON schema whose `tag` / `widget` / `screen` fields are **enums generated per-machine from the canonical model** → the model **physically cannot emit a tag that isn't in the model** (token-level grammar). A **deterministic TS compiler** then re-validates (AJV) against the model, applies ISA-101 rules, resolves bindings, and emits a flat component tree. LLM = proposer; **compiler = authority**. Keep a **no-LLM deterministic default-layout generator** as fallback so the demo never dies (also the "reproducible, same-model→same-HMI" credibility story).

- LLM**: Claude structured outputs `output_config.format=json_schema` (GA) via `@anthropic-ai/sdk` `messages.parse` + `zodOutputFormat`, model `claude-opus-5`. Limits that shape design: **no recursion** → **flat element map** (parent-id + grid slot), **enums only** → per-machine tag/widget enums, **no numeric constraints** → ranges enforced by compiler/AJV. (Optional Vercel AI SDK `generateObject` for model-swap; offline: Ilguidance/xgrammar grammar-constrained decoding.)
- Canonical model (JSON, Zod-typed)**: borrow **OPC UA for Machinery (OPC 40001)** Identification/MachineryItemState + **PackML/ISA-TR88** 17-state machine vocabulary. `Tag{path,dataType,unit,engRange,access,role,semantic,source}` · `Command{writesTag,preconditions,confirm,timeout,expectedFeedback}` · `AlarmDef{tag,condition,priority 1-4,message}`. Load the model "into" the node-opcua server's address space so **"canonical model" and "OPC UA server" are literally the same thing** (strong demo point).
- UI spec + renderer**: `@json-render/core` + `@json-render/react` (flat spec matches the non-recursive LLM schema; streaming render) with a **custom HMI catalog** (ValueTile/Gauge/Trend/StateBadge/AlarmBanner/AlarmList/CommandButton/SetpointInput/PumpSymbol/NavBar), SVG mimics, uPlot trends.
- Size-independence**: proposals carry widget **priority + min-size, not pixels**; compiler solves small(7"/4-col)/medium(10-12"/8-col)/large(12-col) with CSS Grid + **container queries**; below breakpoints drop priority-3 to "More", keep state+alarm banner+E-stop always visible → **"same model, three HMIs on one screen"** demo.
- Sim**: port existing `index.html` physics to fixed-timestep seeded TS; drive PLC logic as a **PackML statechart in XState v5**; expose via **node-opcua server + jsmodbus server** over one in-memory tag table (flip protocol live). Modbus TCP = Schneider-native (M241/ATV630/PM5300).
- PLC-authoritative command contract**: HMI never mutates state; sends `Command{...}` → bridge checks role/preconditions/interlocks/state/range → writes command tag → PLC/sim decides → HMI shows **pending** → **verified/REJECTED(reason)** from feedback tag. Audit every command.

- **Data quality:** OPC UA `DataValue.StatusCode` + timestamps (Good/Uncertain/Bad/stale); Modbus derives quality (timeout→bad, no update >N·poll→stale). Bad/stale = greyed value + last-good ts; **command widgets disabled when feedback not Good**; 10-30 s startup grace.
 - **Change-impact:** build a `graphology` directed graph (tag/command/alarm/interlock/state/widget/screen/test); on `diff(modelV1,V2)` BFS in-edges → classify regenerate / re-validate / re-test / unaffected; regenerate only impacted screens (cache others by hash); visualize impacted subgraph with **sigma.js**; auto-generate **Vitest** cases from command contracts → "validated HMI" = schema-valid + model-consistent + ISA-101-linted + tests pass.
 - **Packaging:** pnpm workspace (model/compiler/bridge/sim/hmi); Vite `vite-plugin-singlefile` for the HMI + one node bridge/sim; **in-process "demo mode"** so `index.html` still runs offline like Round 1.
 - **Team split:** P1 model+compiler+impact-graph+tests · P2 LLM proposer+schema-gen+repair-loop+fallback · P3 sim+node-opcua/jsmodbus+bridge+React catalog.
 - **Top risks/mitigations:** (1) LLM quality/latency → deterministic fallback + pre-cache proposals by model-hash + streaming + warm grammar cache + ISA-101 linter w/ 1 repair round; (2) schema-subset mismatch (no recursion/numerics) → flat map + enums + AJV, test schema-gen day 1; (3) protocol plumbing eats time → `SecurityPolicy.None` (say so) + one tag table + normalized `{value,quality,ts}` frame day 1 + in-process demo mode.
 - **Edge/PS4 future:** compiled tree is renderer-agnostic → **WPE WebKit/Cog** kiosk (unchanged web HMI) or **LVGL 9** (has `lv_web_emscrip`ten browser build) → makes the "size- & renderer-independent" claim credible. **FUXA** (MIT, OPC-UA/Modbus/S7) is the "what exists, non-generative" reference to cite.
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7. SHARPENED POSITIONING & DEMO

One-liner: "We don't generate a screen from a prompt — we generate a validated interface from the machine, and keep it in sync as the machine changes."
Differentiator sentence (safe): "Automatic and even LLM-assisted HMI generation already exist; CANON's contribution is the validation-and-continuity layer — provably-correct bindings against a canonical model, PLC-authoritative consistency, and continuity across regenerations."
5-wow demo: (1) machine context in (OPC UA/AML) → canonical model+graph; (2) NL intent → **validated ISA-101 screen materializes**, widgets tag-bound (hover shows binding+VERIFIED); (3) START → **PLC permissive FALSE** → **REJECTED** (open valve → authorized); (4) "diagnostics for current abnormal condition" → new contextual screen; (5) **PT101 0-10→0-16** → **impact map** → **regenerate** → **regression** → **approve** → **new baseline** (the moment nobody else can show). **Judge red-team answers:** "isn't this MTP/Eigen?" → those are **build-time, static, human-gated**; CANON is **runtime + validated + PLC-authoritative + continuity, and can ingest MTP/AAS**. "how avoid hallucinated tags?" → deterministic compiler validates against canonical model; unverified = UNKNOWN, never rendered as control. **Claims firewall:** aligned-to not certified · advisory AI, PLC controls · SIMULATED clearly labeled · UNKNOWN not fabricated · no "first/novel AI HMI" · no unsourced numbers.